Defence presentation outline

Christophe Rouleau-Desrochers

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1. Content
   (a) Intro, chapter 1, chapter 2, conclusion; will last 15 minutes.
   (b) 1
2. Introduction
   (a) ACC impacts on tree phenology
      i. Spring are earlier
      ii. Autumn delayed
      iii. Lenghtens the GSL
      iv. (Figure): simplified version of conceptual figure?
   (b) Long GS should = more growth
      i. This means longer growing seasons which should increase growth
   (c) This assumed trend is important for carbon models; temperate forest= 40% of global net carbon sink
      i. 1
   (d) Recent research showed that this may not be the case
      i. 1
   (e) Drought and competition as potential growth inhibitors and internal constraints
      i. 1
   (f) Questions: do longer and warmer seasons increase growth in juvenile and mature trees?
      i. 1
   (g) Objectives: answer this question using north american juvenile trees of 4 species from 4 provenances (chapter 1) and mature trees of 11 species (chapter 2)
      i. 1
3. Chapter 1
   (a) Introduction: gs conceptual figure
   (b) Introduction: constraints on growth
   (c) Methods: trees grown from seeds and sourced from 4 provenances; 3.5 degree lat gradient; phenology for 3 years, growth with tree rings
   (d) Methods: bayesian hierarchical models? 4 different predictors
   (e) R/D: More species grew more with warmer seasons than calendar season; because warmer is a better predictor of growth
   (f) R/D: Later EOS increased growth more than SOS; because higher temperature than SOS; suggests looking more at EOS

    (g) R/D: No provenance effect; at least for this small gradient, no evidence of local adaptation on growth
    (h) Conclusion: juvenile trees should grow more, means it would be good for range expansion and forest regenaration

4. Chapter 2
    (a) Introduction: different growth strategies could influence tree growth responses to GS length
    (b) Introduction: fast vs slower growing species
    (c) Introduction: wood porousness and timing of leaf phenology and growth = differences in temperature experienced; also correlates with different life-history strategies
    (d) Introduction: carbon investment in storage may blur annual growth responses
    (e) Methods: citizen science leaf phenology; growth with tree rings at the arboretum
    (f) Methods: stats? Same predictors, but used leaf coloring as EOS.
    (g) R/D: very little effect of season length

5. Conclusion
    (a) 1
    (b) 2